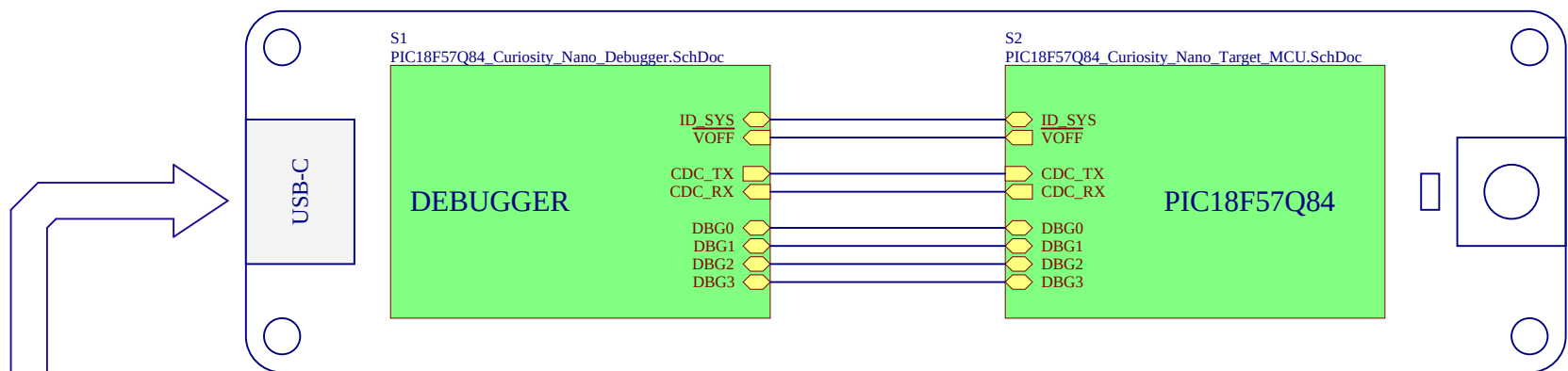
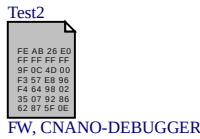
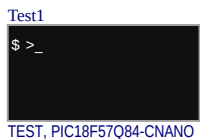
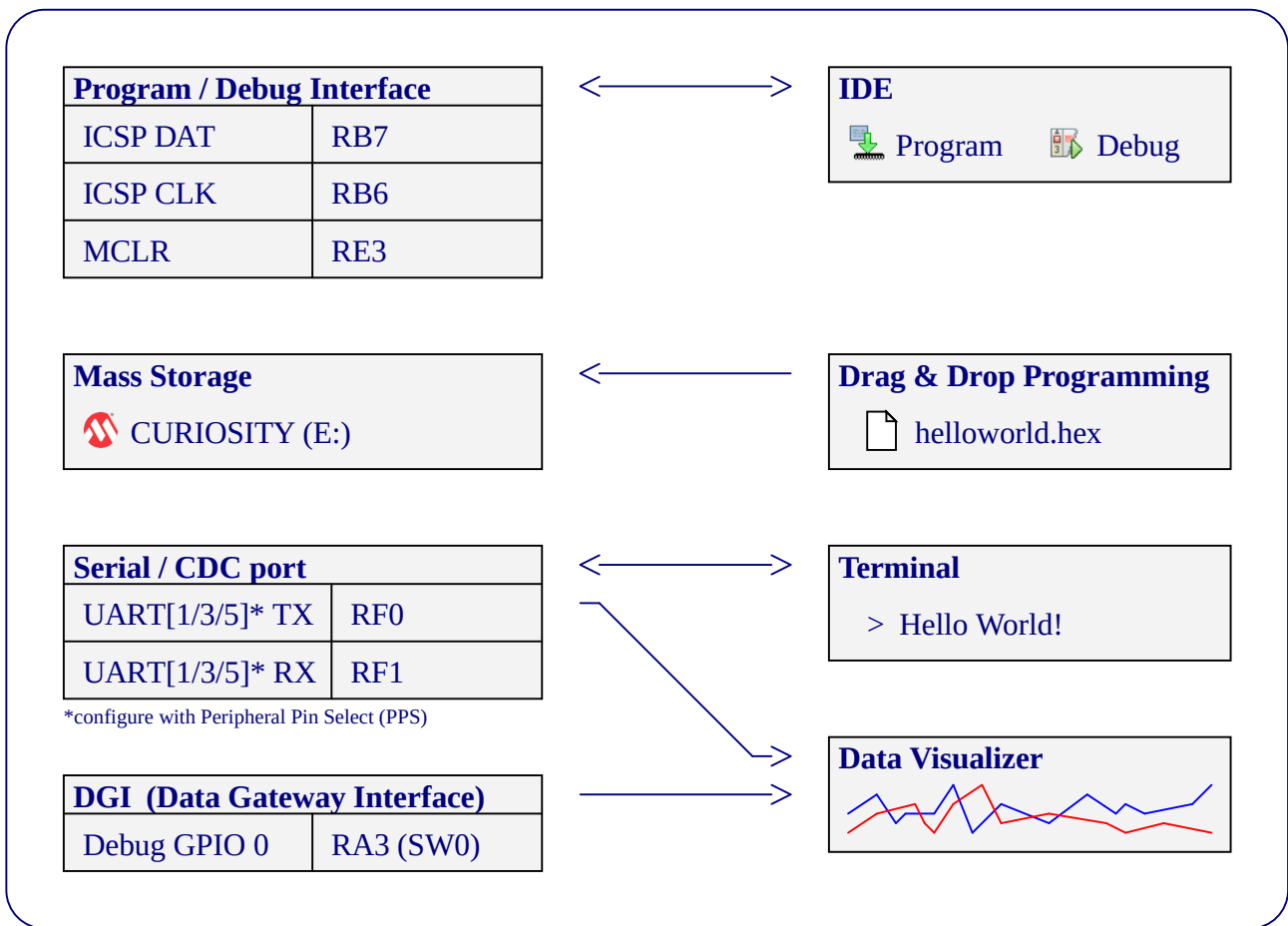


PIC18F57Q84 Curiosity Nano



On-Board Peripherals

LED0	RF3	Active Low
SW0	RA3	Active Low



"Text in Silkscreen on PCB"

CNANO Template Revision: 2.7

Project Owner: PB		 MICROCHIP				
PCB Layout Contact: SLT						
PartNumber: DM182030		Project Title PIC18F57Q84 Curiosity Nano		Variant: Default Assembly		
Sheet Title Top Level						Designed with  Altium.com
Size	A3	SCH #:02-01204	Rev: 6	Date: 18.12.2024		
		PCB #: 04-12268	Rev: 4	Sheet 1 of 4	Altium.com	
File: PIC18F57Q84_Curiosity_Nano_TopLevel.SchDoc						

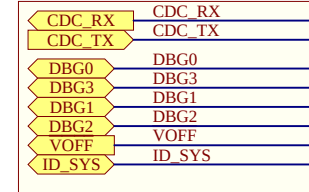
S3
PIC18F57Q84_Curiosity_Nano_Revision_History.SchDoc



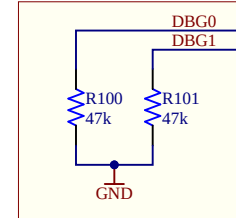
PIC18F57Q84

PIC18F57Q84		
Debugger	Name	Pin
CDC TX	UART RX	RF1
CDC RX	UART TX	RF0
DBG0	ICSPDAT	RB7
DBG1	ICSPCLK	RB6
DBG2	GPI00	RA3
DBG3	MCLR	RE3
VTG	1.8V - 5.5V	

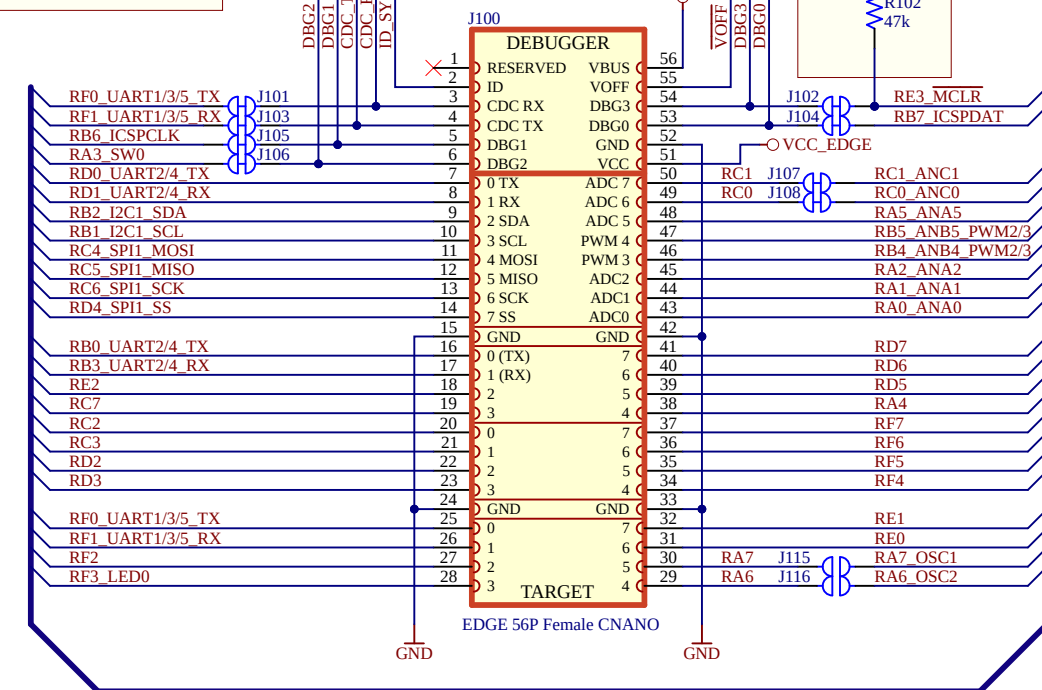
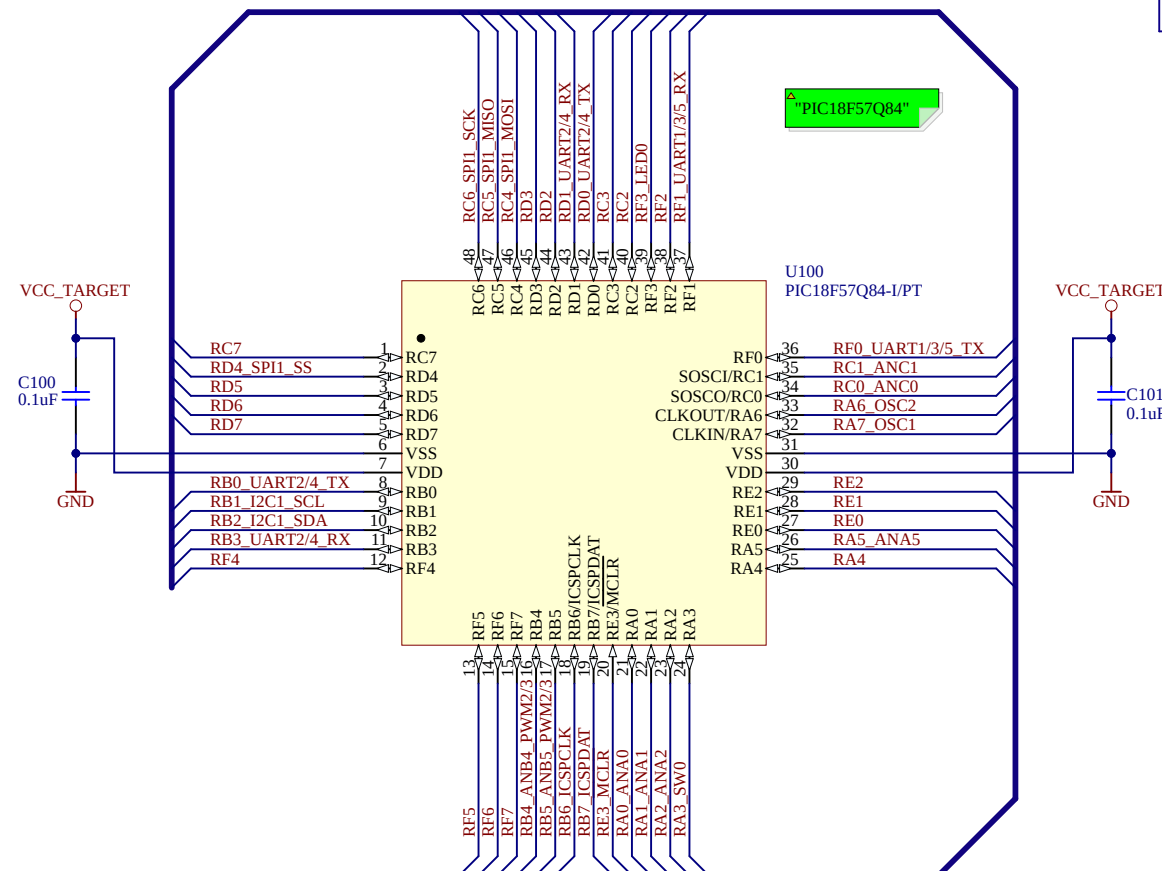
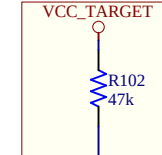
DEBUGGER CONNECTIONS



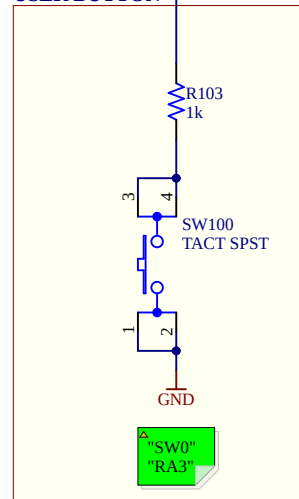
PROG/DEBUG Pull



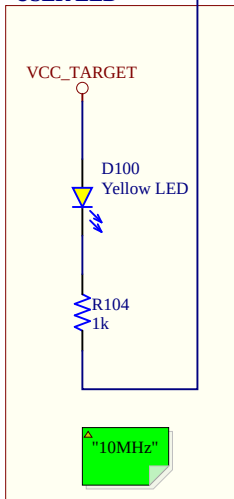
MCLR Pull



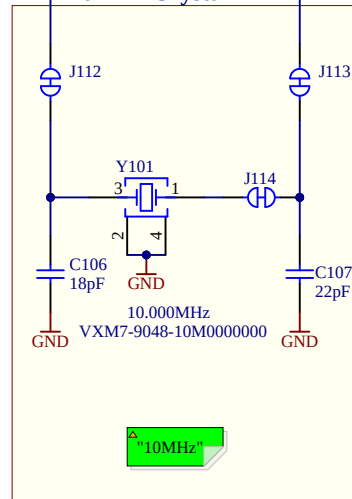
USER BUTTON



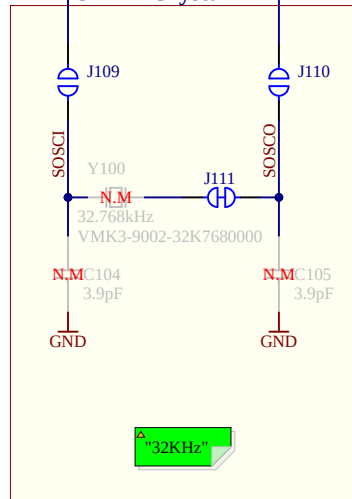
USER LED



10MHz Crystal



32kHz Crystal



10MHz Crystal

Crystal datasheet:
Ccrystal = 12pF
max ESR = 150
Accuracy ±25ppm

Selected in design after verification
C203 = 18pF
C204 = 22pF

Empirical verification of 5 units showed values within +5 ppm clock accuracy.

Crystal safety factor tested by cutting J114 and replacing it with a 750 ohm resistor. The oscillator starts up fine.

Safety Factor = (ESR + R) / ESR
SF = (150 + 750) / 150 = 6

SF >= 5; Excellent

NOTE on UART/CDC:

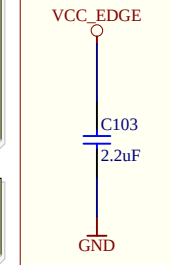
RX/TX on the header denotes the input/output direction of the signal respective to it's source.

CDC TX is output from the DEBUGGER.
CDC RX is input to the DEBUGGER.
TX is output from the TARGET device.
RX is input to the TARGET device.

NOTE on I2C:

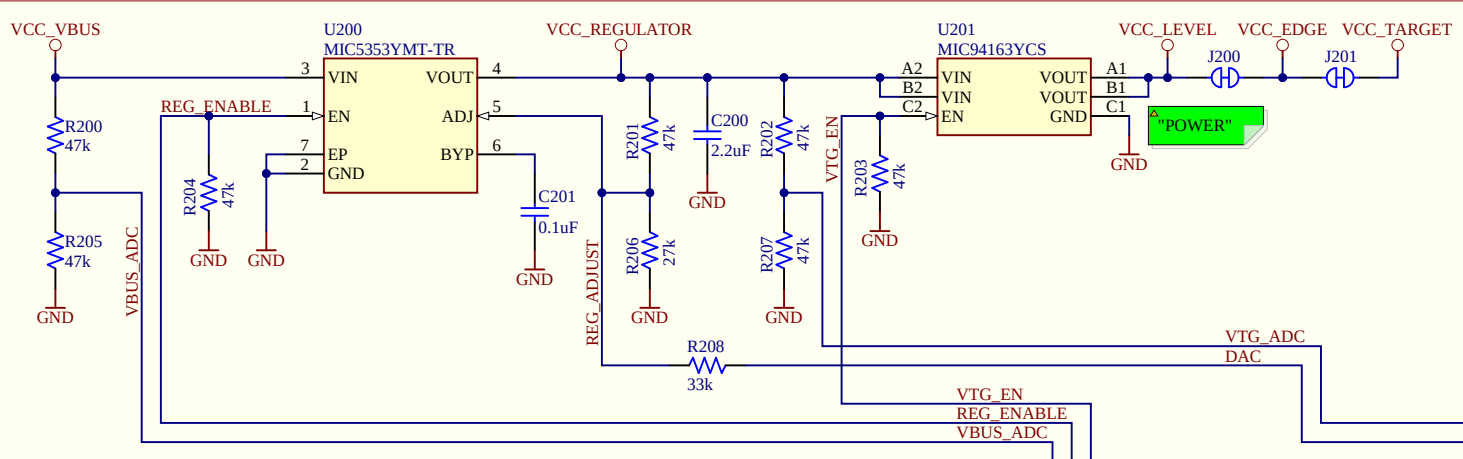
No pull-ups on board. Pull-ups must be mounted close to client device(s).

TARGET BULK



Project Owner: PB		 MICROCHIP			
PCB Layout Contact: SLT					
PartNumber: DM182030	Project Title PIC18F57Q84 Curiosity Nano	Variant: Default Assembly			
Sheet Title Target MCU		Designed with  Altium.com			
Size A3	SCH #: 02-01204			Rev: 6	Date: 18.12.2024
	PCB #: 04-12268			Rev: 4	Sheet 2 of 4
File: PIC18F57Q84_Curiosity_Nano_Target_MCU.SchDoc					

TARGET ADJUSTABLE REGULATOR



J200:
- Cut-strap used for full separation of target power from the level shifters and on-board regulators.
- For current measurements using an external power supply, this strap could be cut for more accurate measurements. Leakage back through the switch is in the micro ampere range.
J201:
- For current measurements using the on-board power supply, this strap must be cut and an ammeter connected across.

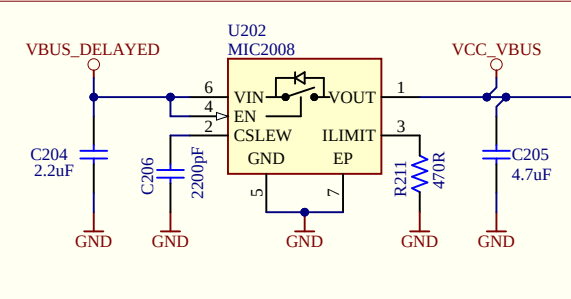
MIC5353:
Vin: 2.6V to 6V
Vout: 1.25V to 5.1V
Imax: 500mA
Dropout (typical): 50mV @ 150mA, 160mV @ 500mA
Accuracy: 2% initial
Thermal shutdown and current limit

Maximum output voltage is limited by the input voltage and the dropout voltage in the regulator. ($V_{max} = V_{in} - \text{dropout}$)

Interface Signal	ICSP™ TARGET	UPDI TARGET	SWD TARGET
CDC TX	UART RX	UART RX	UART RX
CDC RX	UART TX	UART TX	UART TX
DBG0	DAT	UPDI	SWDAT
DBG1	CLK	GPIO	SWCLK
DBG2	GPIO	GPIO	SWO/GPIO
DBG3	MCLR	RESET	RESET
VCC	-	-	-

ADJUSTABLE OUTPUT AND LIMITATIONS:
- The DEBUGGER can adjust the output voltage of the regulator between 1.25V and 5.1V to the target.
- The voltage output is limited by the input (USB), which can vary between 4.40V to 5.25V
- The level shifters have a minimal voltage level of 1.65V and will limit the minimum operating voltage allowed for the target to still allow communication.
- The MIC94163 has a minimal voltage level of 1.70V and will limit the minimum voltage delivered to the target.
- Firmware configuration will limit the voltage range to be within the target specification.

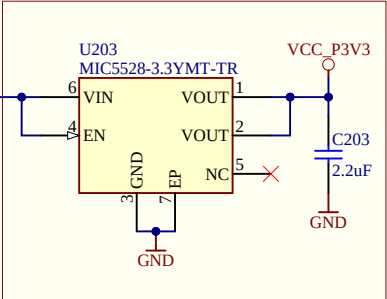
VBUS SLEW RATE- & CURRENT-LIMIT



SLEW RATE LIMIT:
- With C206 = 2200pF, the slew rate of VCC_VBUS is limited to 2 V/ms by the power switch MIC2008.

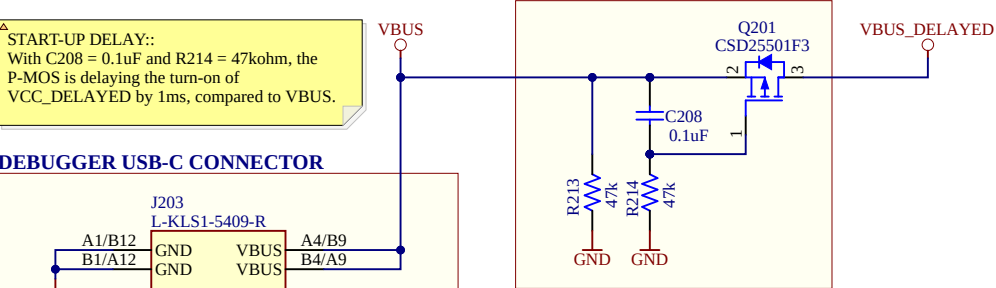
CURRENT LIMIT:
- With R211 = 470ohm, the current through the power switch MIC2008 is limited to 500mA.

DEBUGGER REGULATOR

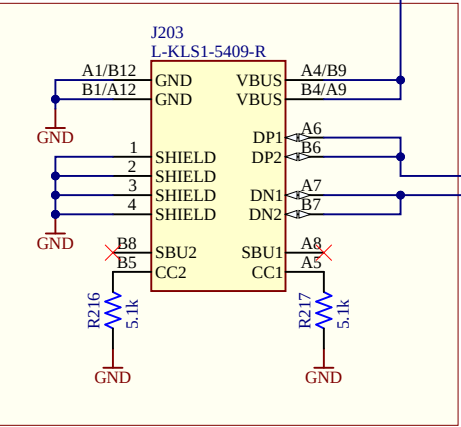


MIC5528:
Vin: 2.5V to 5.5V
Vout: Fixed 3.3V
Imax: 500mA
Dropout: 260mV @ 500mA

VBUS START-UP DELAY

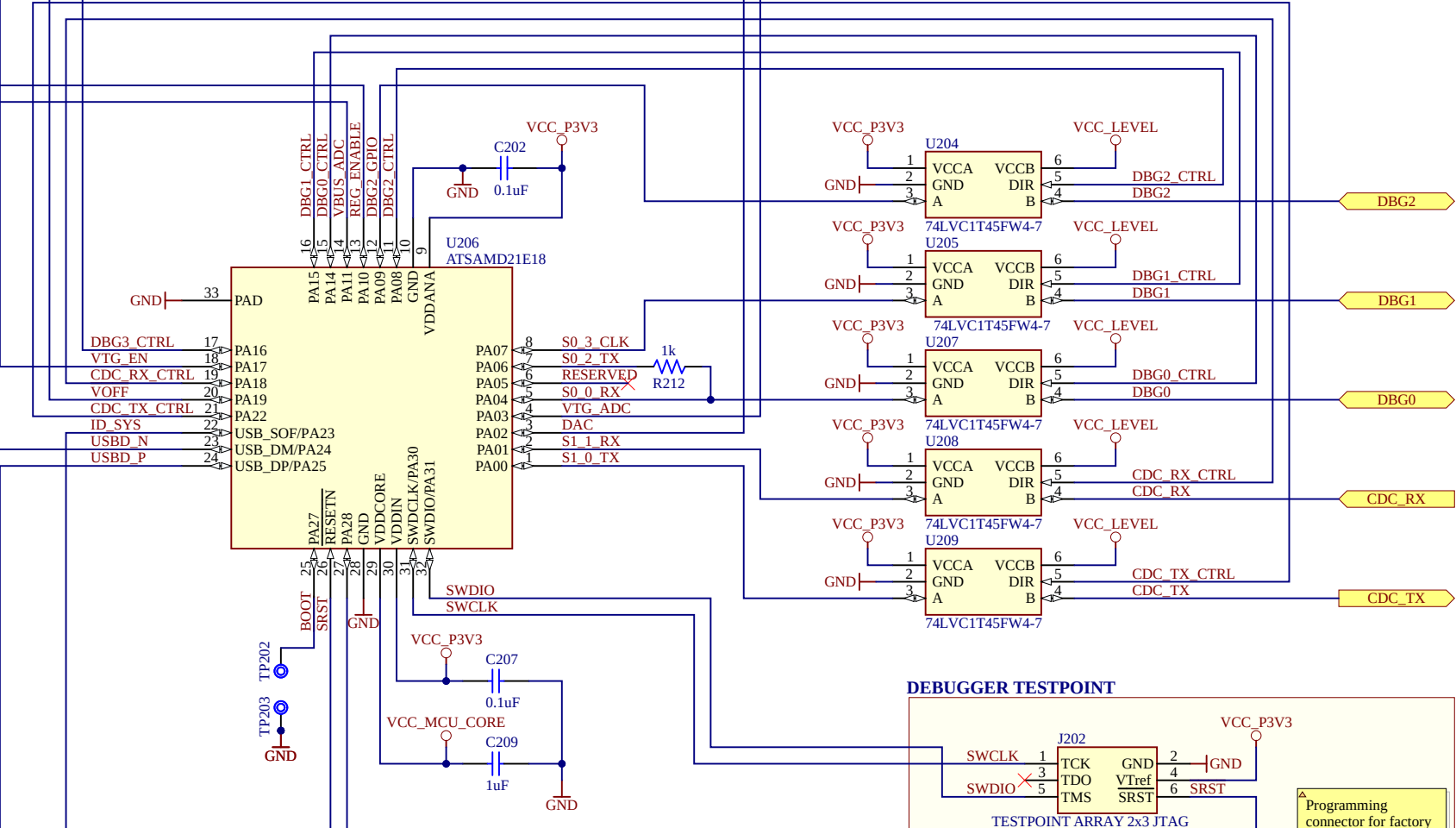


DEBUGGER USB-C CONNECTOR

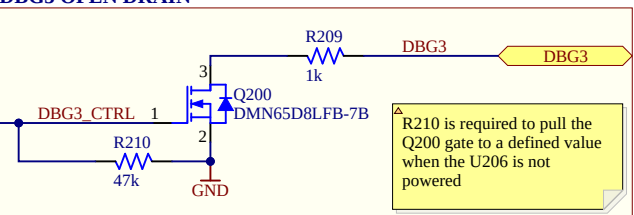


TP200 and TP201 are MTG Holes in the corners of the PCB close to the DEBUGGER section labelled "GND" on the silkscreen.

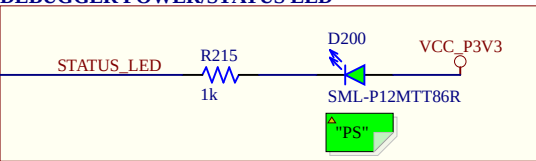
DEBUGGER



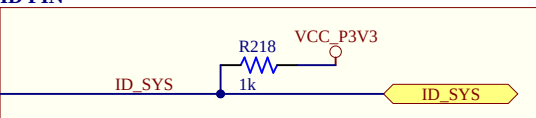
DBG3 OPEN DRAIN



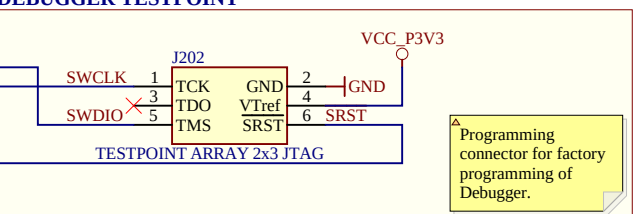
DEBUGGER POWER/STATUS LED



ID PIN



DEBUGGER TESTPOINT



Revision History

PCB Assembly Rev 1:

▲

Design Changes:

Initial Design

PCB:

PCB revision 1

PCB Assembly Rev 2:

▲

Design Changes:

New MCU Pinout

PCB:

PCB revision 2

PCB Assembly Rev 3:

▲

Design Changes:

Added 10MHz crystal
Edge pinout changed to account for 10MHz crystal

PCB:

PCB revision 3

PCB Assembly Rev 4:

▲

Design Changes:

Replaced C203 and C204:
C203: 10pF -> 18pF
C204: 13pF -> 22pF

PCB:

PCB revision 3

PCB Assembly Rev 6:

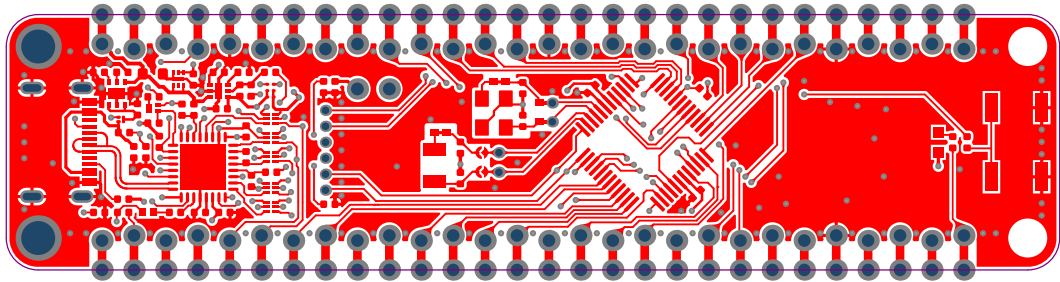
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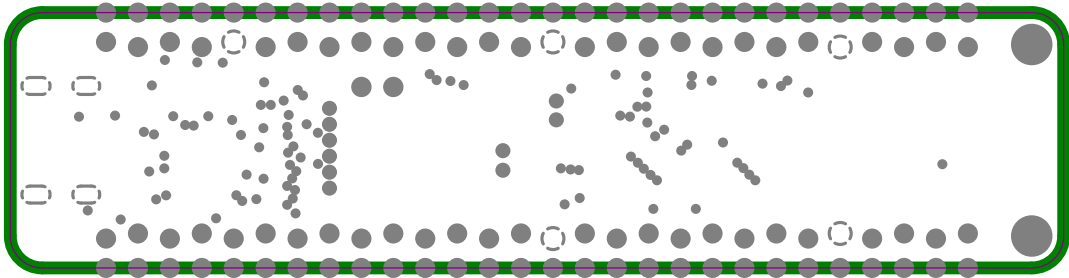
Design Changes:

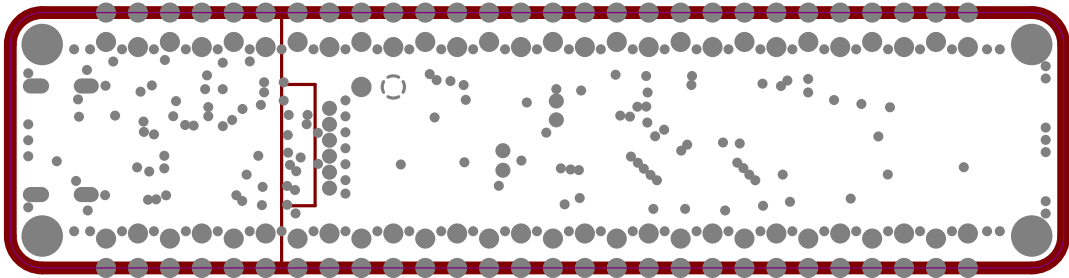
USB-C Connector
Designators reset
Label and switch replaced
Power input and current limit redesign

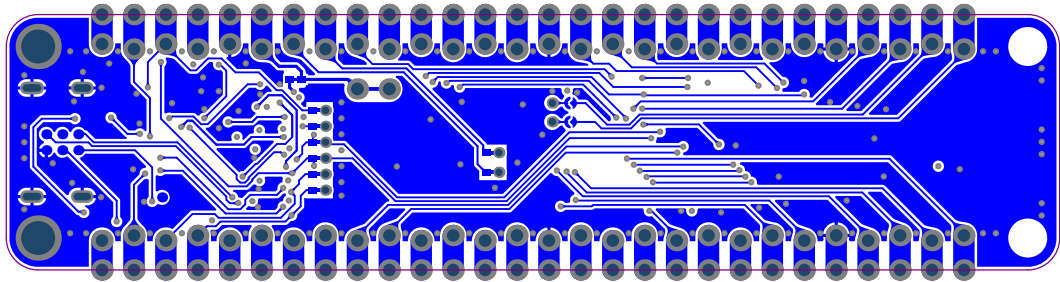
PCB:

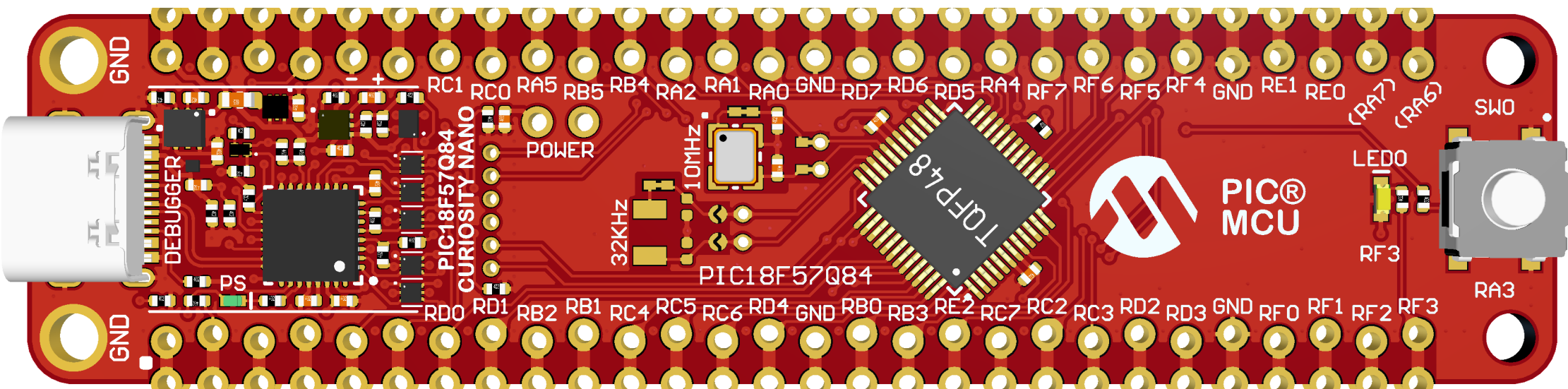
PCB revision 4 - New PCB with USB-C connector













[ASSY# / REV]
SN: [SERIAL]
[DATE yyyy.mm.dd]

04-12268 Rev4
Microchip © 2024

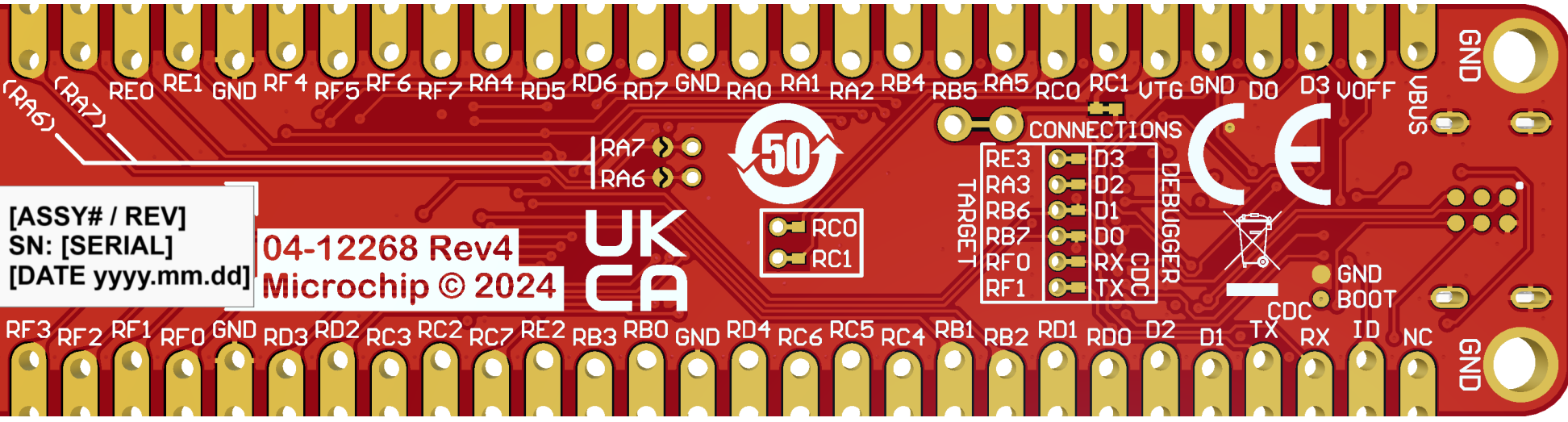
UK
CA

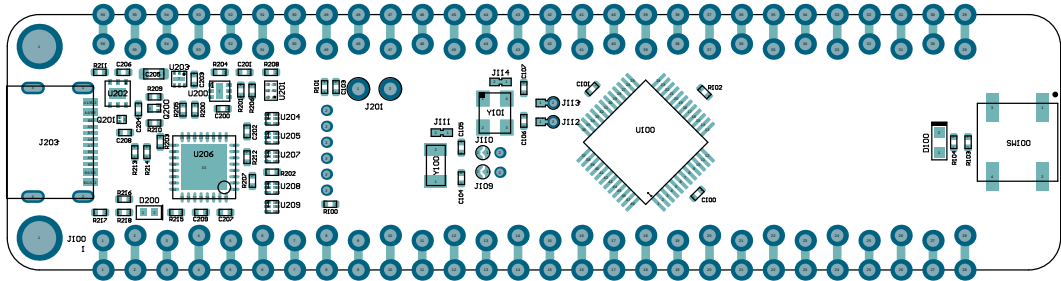


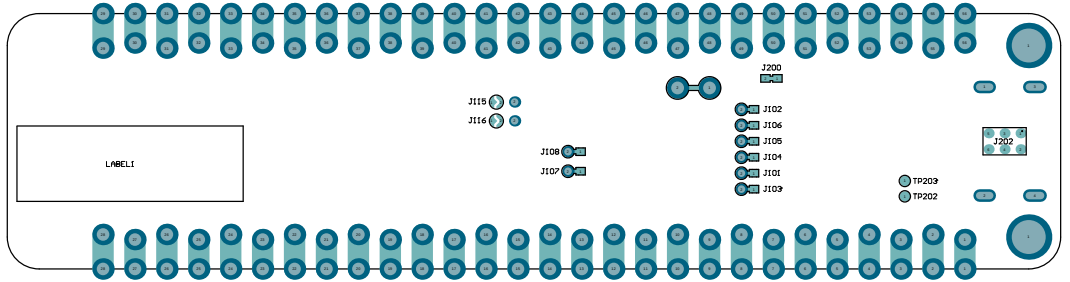
CONNECTIONS			
TARGET	RE3	D3	CDC DEBUGGER
	RA3	D2	
	RB6	D1	
	RB7	D0	
	RF0	RX	
	RF1	TX	



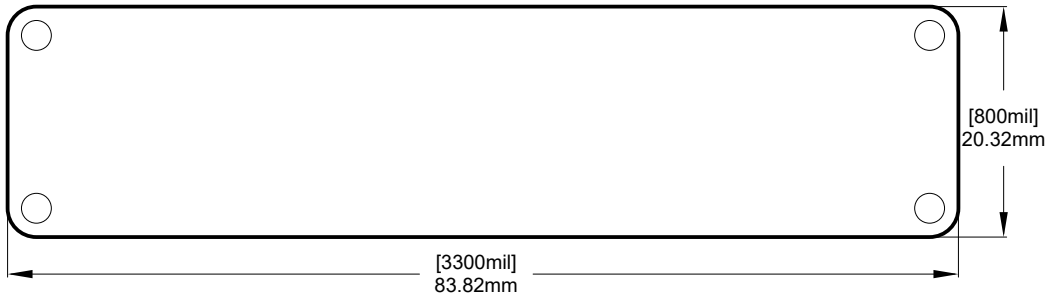
GND
BOOT



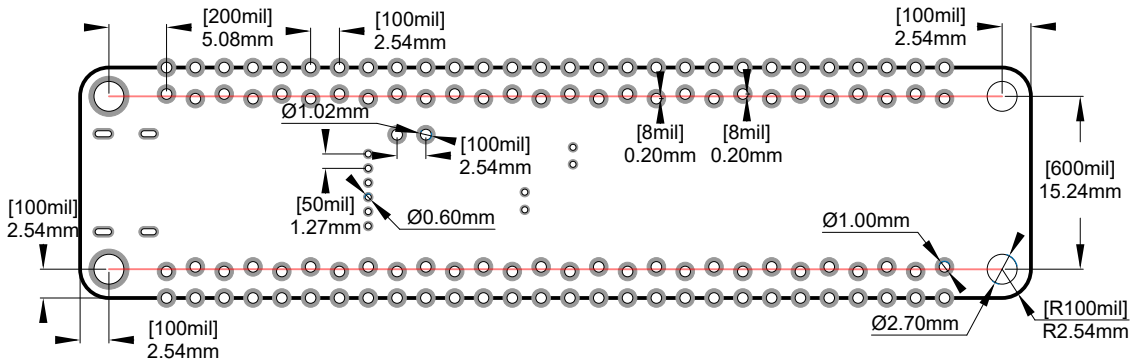




Board Outline

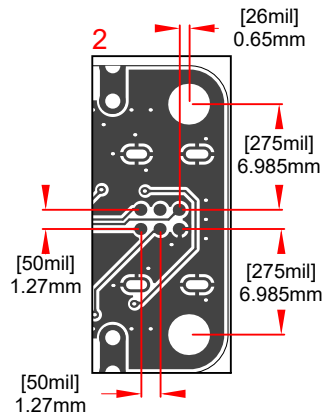
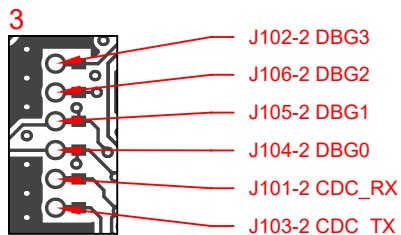
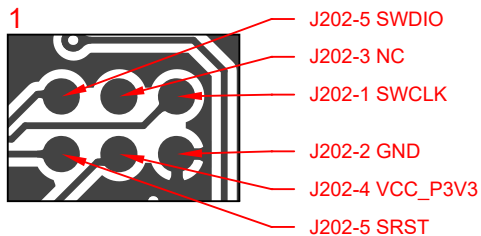
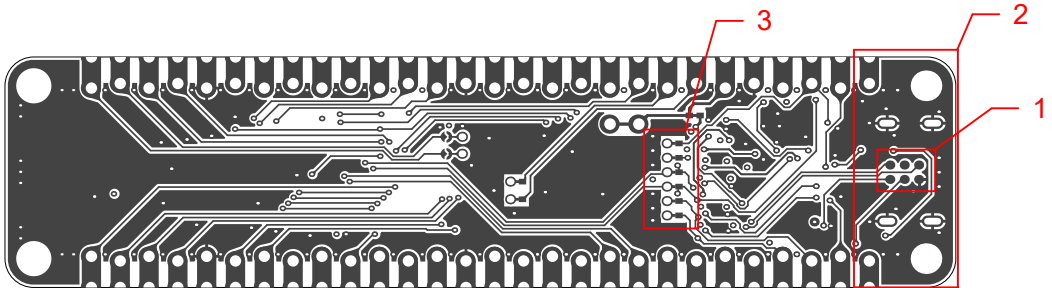
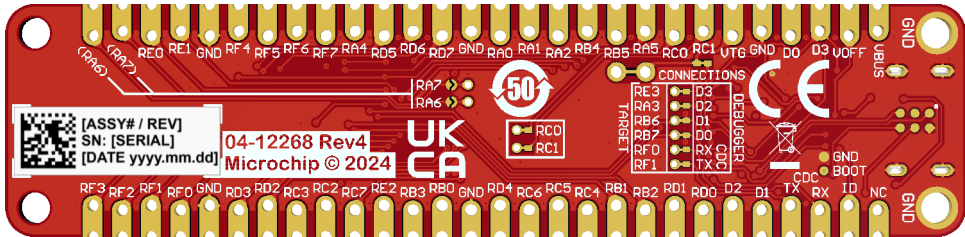


Through-Hole Dimensions



Centerline of staggered through-holes for pin headers.

Test Point Placement [Bottom View]



PCB Stackup

Material	Layer	Thickness	Dielectric Material	Type	Gerber
	Top Overlay			Legend	GTO
Surface Material	Top Solder	0.010mm	Solder Resist	Solder Mask	GTS
Copper	Top Layer	0.035mm		Signal	GTL
Prepreg		0.117mm	PP-016	Dielectric	
Copper	Gnd Plane	0.035mm		Internal Plane	GP1
Core		1.200mm	FR-4	Dielectric	
Copper	Pwr Plane	0.035mm		Internal Plane	GP2
Prepreg		0.117mm	PP-016	Dielectric	
Copper	Bottom Layer	0.035mm		Signal	GBL
Surface Material	Bottom Solder	0.010mm	Solder Resist	Solder Mask	GBS
	Bottom Overlay			Legend	GBO
Total thickness: 1.594mm					

Component list

PIC18F57Q84 Curiosity Nano

PCBA Number: 02-01204

PCBA Revision: 6

Variant: Default Assembly



Print Date: 18.12.2024 10:37:20

Fitted	Designator	Quantity	Value	Manufacturer 1	Manufacturer Part Number 1	Description
Fitted	C100, C101, C201, C202, C207, C208	6	0.1uF	KEMET	C0402C104K4RAC7867	CAP CER 0.1uF 16V 10% X7R SMD 0402
Fitted	C103, C200, C203, C204	4	2.2uF	Murata	GRM155R61A225KE95D	CAP CER 2.2uF 10V 10% X5R SMD 0402
Fitted	C106	1	18pF	Murata	GCM1555C1H180JA16D	CAP CER 18pF 50V 5% COG SMD 0402
Fitted	C107	1	22pF	Murata	GRM1555C1H220JA01D	CAP CER 22pF 50V 5% NPO SMD 0402
Fitted	C205	1	4.7uF	KEMET	C0603C475K8PAC7867	CAP CER 4.7uF 10V 10% X5R SMD 0603
Fitted	C206	1	2200pF	KEMET	C0402C222J3GACTU	CAP CER 2200pF 25V 5% COG SMD 0402
Fitted	C209	1	1uF	KEMET	C0402C105K9PAC7867	CAP CER 1uF 6.3V 10% X5R SMD 0402
Fitted	D100	1	Yellow LED	Rohm Semiconductor	SML-D12Y1WT86	LED YELLOW DIFFUSED 1608 SMD
Fitted	D200	1	LED SINGLE SML-P12MTT86R YELLOW-GREEN	Rohm Semiconductor	SML-P12MTT86R	DIO LED YELLOW-GREEN 2.2V 20mA 25mcd Clear SMD 0402
Fitted	J203	1	USB2.0 Type-C FEMALE	KLS Electronic	L-KLS1-5409-R	CON USB2.0 Type-C Female SMD R/A
Fitted	LABEL1	1	PCBA LABEL 18X6mm	ACT Logimark AS	505462	LABEL PCBA 18x6mm Datamatrix Assy# / Rev / Serial / Date
Fitted	Q200	1	DMN65D8LFB-7B	Diodes Incorporated	DMN65D8LFB-7	TRANS FET N-CH DMN65D8LFB-7B 260mA 60V 430mW X1_DFN1006-3
Fitted	Q201	1	CSD25501F3	Texas Instruments	CSD25501F3	TRANS FET P-CH CSD25501F3 -20V -3.6A 0.076R LGA
Fitted	R100, R101, R102, R200, R201, R202, R203, R204, R205, R207, R210, R213, R214	13	47k	Yageo	RC0402FR-0747KL	RES TKF 47k 1% 1/16W SMD 0402
Fitted	R103, R104, R209, R212, R215, R218	6	1k	Panasonic	ERJ-2RKF1001X	RES TKF 1k 1% 1/10W SMD 0402
Fitted	R206	1	27k	Yageo	RC0402FR-0727KL	RES TKF 27k 1% 1/16W SMD 0402
Fitted	R208	1	33k	Yageo	AC0402FR-0733KL	RES TKF 33k 1% 1/16W SMD 0402
Fitted	R211	1	470R	Yageo	RC0402FR-07470RL	RES TKF 470R 1% 1/16W MF 0402
Fitted	R216, R217	2	5.1k	Panasonic Electronic Components	ERJ-2RKF5101X	RES TKF 5.1k 1% 1/10W SMD 0402
Fitted	SW100	1	TACT SPST	Dailywell Electronics Co.,Ltd.	TS604VM1-035CR-R	SWITCH TACT SPST 12 50mA TS604VM1-035CR-R SMD
Fitted	Test1	1				TEST, PIC18F57Q84-CNANO
Fitted	Test2	1	Debugger Firmware			FW, CNANO-DEBUGGER
Fitted	U100	1	PIC18F57Q84-IPT	Microchip Technology	PIC18F57Q84-IPT	MCHP MCU 8-BIT 64MHz 128kB 8kB PIC18F57Q84-IPT TQFP-48
Fitted	U200	1	MIC5353YMT-TR	Microchip Technology	MIC5353YMT-TR	MCHP ANALOG LDO ADJ 500mA MIC5353YMT-TR MLF-6
Fitted	U201	1	MIC94163YCS	Microchip Technology	MIC94163YCS-TR	IC LOAD SWITCH HI SIDE 3A 6WLCSP
Fitted	U202	1	MIC2008	Microchip Technology	MIC2008YML-TR	MCHP ANALOG POWER SWITCH 5.5V 2.1A MIC2008YML-TR VDFN-6
Fitted	U203	1	MIC5528 3V3	Microchip Technology	MIC5528-3.3YMT-TR	MCHP ANALOG LDO 3.3V MIC5528-3.3YMT-TR 6-TDFN
Fitted	U204, U205, U207, U208, U209	5	74LVC1T45FW4-7	Diodes Incorporated	74LVC1T45FW4-7	IC VOLTAGE TRANSLATOR BI-DIR 1 CIRCUIT 74LVC1T45FW4-7 X2-DFN1010-6
Fitted	U206	1	ATSAMD21E18	Microchip Technology	ATSAMD21E18A-MUT	MCHP MCU 32-BIT 48MHz 256kB 32kB ATSAMD21E18A-MUT QFN-32
Fitted	Y101	1	10.000MHz	Microchip Technology	VXM7-9048-10M0000000	MCHP CRYSTAL 10.000MHz 12pF VXM7-9048-10M0000000 SMD L3.2W2.5H0.8
Not Fitted	C104, C105	0	3.9pF	KYOCERA AVX	04025A3R9CAT2A	CAP CER 3.9pF 50V 0.25pF COG SMD 0402
Not Fitted	Y100	0	32.768kHz	Microchip Technology	VMK3-9002-32K7680000	MCHP CRYSTAL 32.768kHz 7pF VMK3-9002-32K7680000 SMD L3.2W1.5H0.9
		60				
Approved			Notes			